Transfusion Transmissible Infections Among Walk-In Blood Donors at Kisumu Regional Blood Transfusion Centre, Kisumu County, Kenya, 2015

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ABSTRACT

Background: Transfusion transmissible infections (TTIs) are threats to blood safety. This study investigated TTIs among volunteer blood donors at the Kisumu Regional Blood Transfusion Centre (KRBTC) in Kenya.

Methods: We performed a retrospective record review of blood donor registers at KRBTC, 2015. Walk-in donors accepted for donation were analyzed to determine prevalence of human immunodeficiency virus (HIV), hepatitis B (HBV), hepatitis C (HCV), and syphilis. Descriptive and associative statistics were calculated using Microsoft Excel and OpenEpi software.

Results: The records of a total of 3690 walk-in donors were reviewed, and 2046 were included in the analysis. Mean age was 30 ± 9.6 years, 76% male, 51% married, and 80% were Kisumu residents. Seroprevalence of HIV, HBV, HCV, and syphilis was 2.4%, 3.1%, 2.3%, and 1%, respectively.

Conclusions: HBV was the most common TTI among the volunteers. KRBTC should be proactive and address issues of donor self-exclusion, strict adherence to donor selection criteria, and vaccination.

Keywords: transfusion transmissible infections; HIV; hepatitis B; hepatitis C; blood donors; syphilis.

Blood is a vital healthcare resource used in many life-saving medical procedures and simultaneously a potential vector for harmful, sometimes fatal, infectious diseases, such as human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV), and syphilis.[1] Although medical and healthcare institutions have improved their screening of donated blood, transfusion transmissible infections (TTIs) still pose a threat to global blood safety, especially in low- to middle-income countries such as Kenya.[2] With rapid growth of the Kenyan economy in the 21st century, blood transfusion services in Kenya have experienced infrastructure transformation and development. A county-based survey reported Kisumu County to have an HIV prevalence of 19.3% compared to a national prevalence of 5.6%.[3] Despite the high prevalence of HIV in the region, it remains the major source of blood for Kisumu Regional Blood Transfusion Centre (KRBTC).

Consequently, rigorous assessment of suitable prospective donors to protect the safety and sufficiency of the blood supply, safeguarding the health of donors themselves, and ensuring suitable donors are not necessarily deferred, cannot be overemphasized. A global practice of selecting blood donors with low TTI risk followed by effective laboratory screening has reduced the risk of TTIs to very low levels in the past 20 years.[4] The battery of screening tests conducted on donor blood has substantially reduced the risk of TTIs, although it has increased the cost of providing safe blood.[5] While the volume of blood collected has risen due to increasing income and access to care, it has not kept pace with the demand for blood and blood products in Kenya. In this study, we investigated the prevalence of TTIs

Abbreviations

AG, age group; CI, confidence interval; ELISA, enzyme-linked immunosorbent assay; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; KRBTC, Kisumu Regional Blood Transfusion Centre; OR, odds ratio; TTI, transfusion-transmissible infection.

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Table 1. Distribution of Transfusion Transmitted Infection by Sociodemographic Characteristics at Kisumu Regional Blood Transfusion Centre, 2015 (n = 2046[a])

Age (Years)	Population (%)	HIV (%)	HBV (%)	HCV (%)	Syphilis (%)	Total (%)
<20	156 (8.8)	1 (0.6)	3 (1.9)	6 (3.8)	4 (2.6)	14 (2.2)
20-29	813 (46)	20 (2.5)	29 (3.6)	24 (3)	8 (1)	81 (2.5)
30-39	508 (28.7)	21 (4.1)	20 (3.9)	10 (2)	6 (1.2)	57 (2.8)
40-49	198 (11.2)	6 (3.0)	7 (3.5)	7 (3.5)	0 (0)	20 (2.5)
50-59	82 (4.6)	3 (3.7)	4 (4.9)	1 (1.2)	2 (2.4)	10 (3)
≥60	10 (0.6)	0 (0)	1 (10)	0 (0)	0 (0)	1 (2.5)
Total	1767 (100)	51 (2.9)	64 (2.7)	48 (1.1)	20 (1.1)	183 (2.6)
Gender						
Male	1360 (76.1)	35 (2.6)	54 (4)	41 (3)	16 (1.2)	146 (10.7)
Female	427 (23.9)	16 (3.7)	13 (3)	9 (2.1)	6 (1.4)	44 (10.3)
Total	1787 (100)	51 (2.9)	67 (3.7)	50 (2.8)	22 (1.2)	190 (10.6)
County						
Kisumu	1255 (80.3)	38 (3.0)	45 (3.6)	38 (3)	16 (10.9)	137 (10.9)
Siaya	125 (8.0)	6 (4.8)	9 (7.2)	4 (3.2)	1 (0.8)	20 (16)
Homabay	67 (4.3)	1 (1.5)	5 (7.5)	1 (1.5)	3 (4.5)	10 (14.9)
Vihiga	58 (3.7)	1 (1.7)	2 (3.4)	3 (5.2)	1 (1.7)	7 (12.1)
Nairobi	32 (2.0)	1 (3.1)	1 (3.1)	2 (6.3)	0 (0)	4 (12.5)
Kakamega	26 (1.7)	0 (0)	2 (7.7)	0 (0)	0 (0)	2 (7.7)
Total	1563 (100)	47 (3.0)	64 (4.1)	48 (3.1)	21 (1.3)	180 (11.5)
Education						
None	2 (0.1)	0	0	0	0	0
Primary	307 (19)	14 (4.6)	19 (6.2)	19 (6.2)	9 (2.9)	61 (19.9)
Secondary	568 (35.2)	21 (3.7)	17 (3)	18 (3.2)	8 (1.4)	64 11.3)
Tertiary	738 (45.7)	14 (1.9)	22 (3)	10 (1.4)	5 (0.7)	51 (6.9)
Total	1615 (100)	49 (3)	58 (3.6)	47 (2.9)	22 (1.4)	176 (10.7)

[a]*Any record that did not contain the specific part of the donor's sociodemographic information was excluded from the analysis, therefore denominators differ from one variable to the next.*
HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus.

among walk-in volunteer blood donors at a large regional blood donation facility.

Materials and Methods

Study Site

This study was conducted at KRBTC. KRBTC is the principal supplier of blood and blood products to 62 facilities in the multicounty region of western Kenya. A blood bank technical officer collects venous blood from all eligible donors after they complete a physical examination. The venous blood collected is tested for HIV, HBV, HCV, and syphilis.

Study Design

This was a retrospective review of blood donation records from walk-in volunteers who presented for blood donation at KRBTC during the year 2015.

Data Source

We reviewed electronic and manual registers containing records of blood donors who reported to KRBTC from January 1 to December 31, 2015. KRBTC maintains donor questionnaire forms, records with details of donors, and the results of serological tests.

Laboratory Methods

All blood tests were done at KRBTC laboratory. The first phase of the screening process of all volunteers is via questionnaire. Technicians also measure blood pressure, pulse rate, weight, and hemoglobin levels. Those eligible for donation after this first phase have blood drawn for additional screening. Routine testing for HIV1 and HIV2 is with Vironostika Uniform II Ag/Ab fourth-generation HIV test (bioMerieux SA, Marcy-l'Étoile, France). Testing for HBV and HCV were via DIAQUICK (DIALAB GmbH, Wiener Neudorf, Austria) rapid tests and standard enzyme-linked immunosorbent assays (ELISAs) for confirmation of positive results. Syphilis testing was with the rapid plasma reagin carbon technique.

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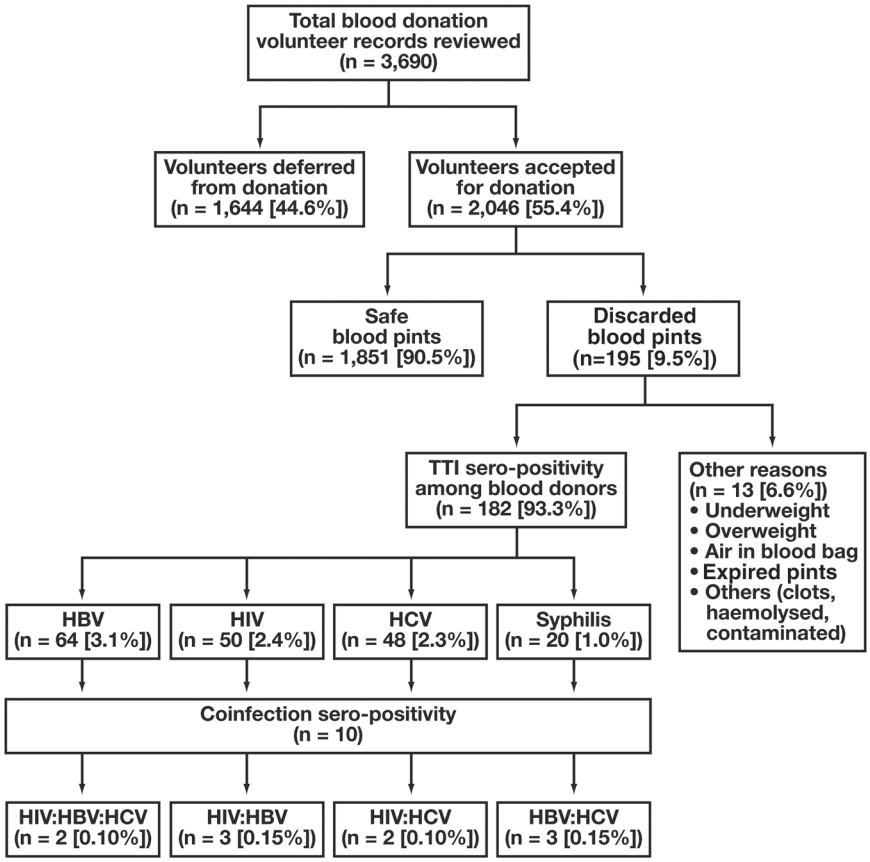


Figure 1

Summary of total blood records reviewed and transfusion transmissible infections (TTIs), Kisumu Regional Blood Transfusion Center (KRBTC), Kenya, 2015.

Exclusion Criteria

Any record that did not contain the blood donor's sociodemographic information was excluded from the analysis for the missing variable.

Statistical Analysis

All data were entered into Microsoft Excel to calculate descriptive statistics. We used Open-Epi to calculate associative statistics and 95% confidence intervals.

Ethical Considerations

Study approval was granted by the management and head of KRBTC. Regarding confidentiality, no names were involved in data collection and analysis processes as only codes were used to identify donors. Blood donor consent was not needed for this study because it did not directly involve volunteers, but rather donation units that were not linked to donor identities.

Results

The records of a total of 3690 walk-in volunteers were reviewed, with 1644 (44.6%) volunteers deferred from donating, leaving a total of 2046 (55.6%) included in the analysis (**Table 1**). The main reason for deferral (n = 418 [25.4%]) from donation was due to low hemoglobin levels as indicated by the blood sample testing. A total of 2046 (55.4%) donors were accepted to donate blood. Mean age was 30 ± 9.6 years, 1055 (56.3%) were first-time donors,

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Table 2. Bivariate Analyses of the Blood Donors' Transfusion Transmissible Infection (TTI) Socio-demographic Variables, Kisumu Regional Blood Transfusion Centre, Kenya, 2015

Variables	2x2 Table Components	Odds Ratio	95% Confidence Interval
Age Group (AG)	<20/>20	0.8	0.5-1.5
	20-29/all AG	0.9	0.7-1.3
	30-39/all AG	1.14	0.8-1.6
	40-49/all AG	0.97	0.6-1.6
	50-59/all AG	1.2	0.6-2.4
	>60/all AG	0.96	0.12-7.63
Gender	Male/female	1.0	0.73-1.49
Level of Education	Primary/secondary	4.9	3.3-7.0
	Primary/tertiary	6.1	4.1-9.1
	Secondary/tertiary	2.5	1.7-3.7
	Primary/Postprimary	2.6	1.8-3.6
	Tertiary/below tertiary	0.4	0.3-0.6
Marital Status	Single	0.75	0.55-1.02
	Married	1.29	0.95-1.76
	Widowed	2.61	0.71-9.57
	Divorced	0.78	0.10-6.09
Donation Frequency	New/old donors	1.74	1.27-2.38
TTIs	HIV/HBV	2.0	0.6-6.7
	HIV/HCV	1.8	0.4-7.5
	HBV/HCV	2.1	0.6-7.0

AG, age group; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus.

and 1360 (76.0%) were men. Among donors, 1255 (80.3%) were from Kisumu County, 891 (51.2%) were married, and 738 (46.1%) had at least a tertiary education.

Of 2046 donors, 195 (9.5%) of the blood donations were deemed unsafe and then discarded, of which 182 (93.3%) tested positive for at least 1 TTI, and 13 (6.6%) were discarded for other reasons (underweight, overweight, air in blood bag, clots, or other contamination). Seroprevalence of HIV, HBV HCV, and syphilis among donors was 2.4% (n = 50), 3.1% (n = 64), 2.3% (n = 48), and 1% (n = 20), respectively. Two (0.10%) donors were positive for HIV, HBV, and HCV; 3 (0.15%) were positive for HIV and HBV; 2 (0.10%) were positive for HIV and HCV; and 3 (0.15%) were positive for HBV and HCV (**Figure 1**).

Bivariate analyses showed that those who were first-time blood donors were 1.7 times more likely to test positive for at least 1 TTI compared to non-first-time donors (OR = 1.7, 95% CI 1.27-2.38). Likewise, those with only primary-level education were 6.1 times more likely to test positive for at least 1TTI compared to those with higher levels of education (OR = 6.1, 95% CI 4.1-9.1). Those with a tertiary education were 40% less likely to test positive for at least 1 TTI (OR = 0.4, 95% CI 0.3-0.6). The results of bivariate analyses by age group and marital status were statistically insignificant, as all confidence intervals included 1, thus not allowing us to draw conclusions about the effects those variables may have had on the outcome (**Table 2**).

Discussion

Transfusion of blood and blood products is a life-saving measure and helps innumerable people worldwide. However, blood transfusion is also an important mode of transmission of infection. This study showed that most (76%) blood donors were young men. This may be attributed to deferral of potential female donors due to anaemia, pregnancy, breastfeeding, or recent childbirth, which are all criteria for donor exclusion. Multiple studies in Africa report a male dominance in blood donation programs.[6] In the African context, these studies show a general belief that men are healthier than women, which may explain the tendency for more men to volunteer to donate blood. The mean age of the donors was 30 ± 9.6 years, comprised mostly of high school and college students, as these groups were more willing to donate blood. These age-related findings of our current study are consistent with other studies, where more than half of the donors were in the age group of 17 to 25 years.[7] In this study, 46% of donors had attained at least a secondary school education. These results are in line with other African studies that showed that volunteer donors had higher educational levels than blood banks, whose donors are largely familial.[7] Comparison of seroprevalence of TTIs by sex of blood donors in this study may not be valid due to the high percentage of male blood donors. This is because most women were excluded from donation due to medical deferrals (eg, low hemoglobin levels). The sex-related analysis of the viral carriage of HBV from multiple studies showed that men were more often infected than women, with a prevalence of 14.7% and 6.4%, respectively.[8] Our data showed that men were more often positive for HCV antibodies, with a seroprevalence of 3%. However, HIV was more prevalent in women than men. The prevalence of HIV in KRBTC was higher in women (3.7%) than men (2.6%), though this difference was not statistically significant. This is unlike what was reported in Osogbo,

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Southwest Nigeria, in which more men (82%), were infected by HIV than women (18%).[9] The HIV seropositivity among female voluntary donors in this study is possibly because women of all ages are more likely than men to become infected with HIV, which studies have shown may be attributed to socioeconomic, cultural, and biological factors that contribute to women's increased vulnerability to HIV.[10]

Conclusions

We collected data on the seroprevalence of TTIs among walk-in blood donors at a large blood bank in Kisumu, Kenya. Our study results show that TTIs remain a threat to the safety of Kenya's blood supply, and more sensitive screening methods and more targeted recruitment of regular donors are still important practices needed to improve blood safety and availability. **LM**

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